

Data 100, Summer 2026

Discussion #8 Solutions

Note: Your TA will probably not cover all the problems. This is totally fine; the discussion worksheets are not designed to be finished within an hour. They are deliberately made slightly longer so they can serve as resources you can use to practice, reinforce, and build upon concepts discussed in lectures, labs, and homework.

One-Hot Encoding

In order to include a qualitative variable in a model, we convert it into a collection of boolean vectors that only contain the values 0 and 1. For example, suppose we have a qualitative variable with 3 possible values, call them A , B , and C , respectively. For concreteness, we use a specific example with 10 observations:

$$[A, A, A, A, B, B, B, C, C, C]$$

We can represent this qualitative variable with 3 boolean vectors that take on values 1 or 0 depending on the value of this qualitative variable. Specifically, the values of these 3 boolean vectors for this dataset are **indicator variables** $\vec{x}_A$, $\vec{x}_B$, and $\vec{x}_C$ such that its entries are:

$$x_{i,k} = \begin{cases} 1 & \text{if } i\text{-th observation has value } k \in \{A, B, C\} \\ 0 & \text{otherwise.} \end{cases}$$

Let y represent any vector of outcome variables, and y_i is the value of said outcome for the i -th subject. This representation is also called **one-hot encoding**.

$$\mathbb{X} = \begin{bmatrix} | & | & | \\ \vec{x}_A & \vec{x}_B & \vec{x}_C \\ | & | & | \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

In this discussion, we will show that the fitted coefficients for $\vec{x}_A$, $\vec{x}_B$, and $\vec{x}_C$ are $\bar{y}_A$, $\bar{y}_B$, and $\bar{y}_C$ (the average of the y_i values for each of the groups, respectively).

1. (a) Show that if you augment your $\mathbb{X}$ matrix with an additional $\vec{1}$ bias vector as shown below, $\mathbb{X}^T \mathbb{X}$ is not full rank. Conclude that the new $\mathbb{X}^T \mathbb{X}$ is not invertible, and we cannot use the least squares estimate in this situation.

$$\mathbb{X} = \begin{bmatrix} | & | & | & | \\ \vec{1} & \vec{x}_A & \vec{x}_B & \vec{x}_C \\ | & | & | & | \end{bmatrix}$$

Solution: Solution 1:

By the definition of one-hot encoding, the one-hot-encoded columns of $\mathbb{X}$ sum up to $\vec{1}$:

$$\vec{x}_A + \vec{x}_B + \vec{x}_C = \vec{1}$$

Since the leftmost vector of $\mathbb{X}$ is a linear combination of the other vectors, $\mathbb{X}$ is not full column rank. Since $\mathbb{X}$ has the same rank as $\mathbb{X}^T \mathbb{X}$ (proof is out of scope), $\mathbb{X}^T \mathbb{X}$ is not invertible.

Solution 2:

We can show that $\mathbb{X}^T \mathbb{X}$ is equal to the following.

$$\mathbb{X}^T \mathbb{X} = \begin{bmatrix} n & n_A & n_B & n_C \\ n_A & n_A & 0 & 0 \\ n_B & 0 & n_B & 0 \\ n_C & 0 & 0 & n_C \end{bmatrix}$$

It can be observed that since $n_A + n_B + n_C = n$, the sum of the final 3 columns subtracted from the first column yields the zero vector $\vec{0}$. By the definition of linear dependence, we can conclude that this matrix is not full rank, and hence, we cannot invert it. As a result, we cannot compute our least squares estimate since it requires $\mathbb{X}^T \mathbb{X}$.

- (b) Show that the columns of $\mathbb{X}$ (without the additional $\vec{1}$ bias vector) are orthogonal.

Solution: The argument is the same for any pair of $\mathbb{X}$'s columns so we show the orthogonality for one pair, $\vec{x}_A \cdot \vec{x}_B$.

$$\begin{aligned} \vec{x}_A \cdot \vec{x}_B &= \sum_{i=1}^{10} x_{A,i} x_{B,i} \\ &= \sum_{i=1}^4 (1 \times 0) + \sum_{i=5}^7 (0 \times 1) + \sum_{i=8}^{10} (0 \times 0) \\ &= 0 \end{aligned}$$

- (c) Show that

$$\mathbb{X}^T \mathbb{X} = \begin{bmatrix} n_A & 0 & 0 \\ 0 & n_B & 0 \\ 0 & 0 & n_C \end{bmatrix}$$

where n_A , n_B , and n_C are number of observations in group A , B , or C .

Solution: Here, we note that

$$\mathbb{X}^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

We also note that

$$\mathbb{X}^T \mathbb{X} = \begin{bmatrix} \vec{x}_A^T \vec{x}_A & \vec{x}_A^T \vec{x}_B & \vec{x}_A^T \vec{x}_C \\ \vec{x}_B^T \vec{x}_A & \vec{x}_B^T \vec{x}_B & \vec{x}_B^T \vec{x}_C \\ \vec{x}_C^T \vec{x}_A & \vec{x}_C^T \vec{x}_B & \vec{x}_C^T \vec{x}_C \end{bmatrix}$$

Since we earlier established the orthogonality of the vectors in $\mathbb{X}$, we find $\mathbb{X}^T \mathbb{X}$ to be the diagonal matrix:

$$\mathbb{X}^T \mathbb{X} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} n_A & 0 & 0 \\ 0 & n_B & 0 \\ 0 & 0 & n_C \end{bmatrix}$$

(d) Show that

$$\mathbb{X}^T \mathbb{Y} = \begin{bmatrix} \sum_{i \in A} y_i \\ \sum_{i \in B} y_i \\ \sum_{i \in C} y_i \end{bmatrix}$$

where i is an element in group A, B , or C .

Solution: Note in the previous solution we found $\mathbb{X}^T$. The solution follows from recognizing that for a row in $\mathbb{X}^T$, e.g., the first row, we have

$$\sum_{i=1}^{10} x_{A,i} \times y_i = \sum_{i=1}^4 y_i = \sum_{i \in \text{group A}} y_i$$

(e) Use the results from the previous questions to solve the normal equations for $\hat{\theta}$, i.e.,

$$\begin{aligned} \hat{\theta} &= [\mathbb{X}^T \mathbb{X}]^{-1} \mathbb{X}^T \mathbb{Y} \\ &= \begin{bmatrix} \bar{y}_A \\ \bar{y}_B \\ \bar{y}_C \end{bmatrix} \end{aligned}$$

Solution: By inspection, we can find

$$[\mathbb{X}^T \mathbb{X}]^{-1} = \begin{bmatrix} \frac{1}{n_A} & 0 & 0 \\ 0 & \frac{1}{n_B} & 0 \\ 0 & 0 & \frac{1}{n_C} \end{bmatrix}$$

When we pre-multiply $\mathbb{X}^T \mathbb{Y}$ by this matrix, we get

$$\begin{bmatrix} \frac{1}{n_A} & 0 & 0 \\ 0 & \frac{1}{n_B} & 0 \\ 0 & 0 & \frac{1}{n_C} \end{bmatrix} \begin{bmatrix} \sum_{i \in A} y_i \\ \sum_{i \in B} y_i \\ \sum_{i \in C} y_i \end{bmatrix} = \begin{bmatrix} \bar{y}_A \\ \bar{y}_B \\ \bar{y}_C \end{bmatrix}$$

Human Contexts and Ethics: Case Study

2. Which of the following scenarios strike you as unfair and why? You can choose more than one. There is no single right answer, but you must explain your reasoning.
 - A. A homeowner whose home is assessed at a higher price than it would sell for.
 - B. A homeowner whose home is assessed at a lower price than it would sell for.
 - C. An assessment process that systematically overvalues inexpensive properties and undervalues expensive properties.
 - D. An assessment process that systematically undervalues inexpensive properties and overvalues expensive properties.

Solution: Your findings from this discussion can be used for Project A2, so we won't release solutions to this question.

Here are some discussion points:

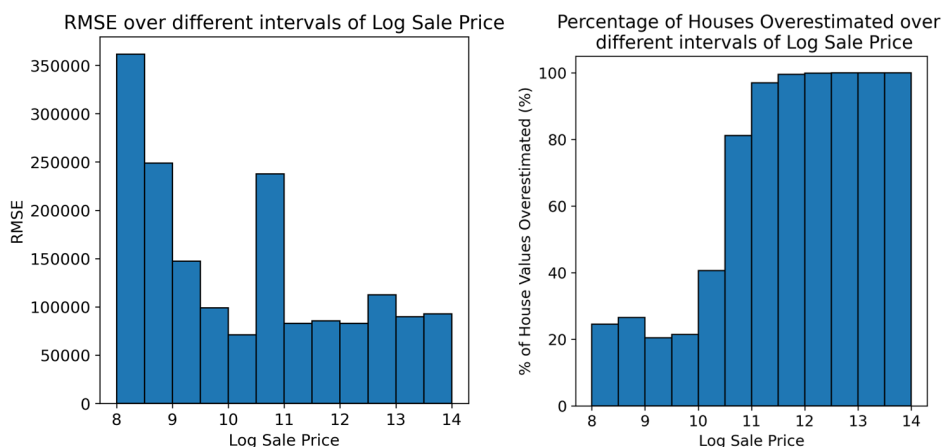
- What is the definition of “fairness” in this context?
- Which individual or group of people are considered when we evaluate fairness?
- Are we talking about whether each situation above is fair to an individual homeowner, a subgroup of society, or the whole society?
- At the level of the individual homeowner, any error can feel unfair: Why am *I* getting over-/under-taxed?
- At the level of the society, is it fair to have a general over or underestimation that varies with the value of the home (a [regressive](#) or [progressive](#) tax scheme)?

Other important things to note:

- Some may feel that both regressive and progressive are unfair; that any systematic errors should not depend at all on the value of the property

- It is possible to have a tax scheme that is systematically fair tax (i.e., not regressive or progressive) but is still quite inaccurate and still generates errors for individual homeowners. However, a tax scheme that generates no errors for homeowners will also have no systematic unfairness.
- Unfairness of the second type (scenarios C and D) is a form of statistical bias that varies as a function of y (the response variable).

3. Suppose you created a model to predict property values (as you are doing presently!) and wanted to understand whether your model's predictions align more with Scenario C or D from the last question. You decide to break down your data into different intervals depending on the true Log Sale Price and compute the Root Mean Squared Error (RMSE) and percentage of property values overestimated for each interval. Using this information, you create the plots below:



Which plot would be more useful in determining whether the model's predicted property values align more with Scenario C or D? Provide a brief justification for your answer.

Solution: Your findings from this discussion can be used for Project A2, so we won't release solutions to this question. A good starting point to consider when answering the question: How does the sign of the residual relate to whether a property is overvalued or undervalued?

More broadly, when answering the written questions at the end of A2, you may also consider the following questions: Does a low RMSE necessarily mean a model's predictions are more accurate? Does a low RMSE necessarily mean a model's predictions are more "fair"? What is the difference between your answers to both questions?